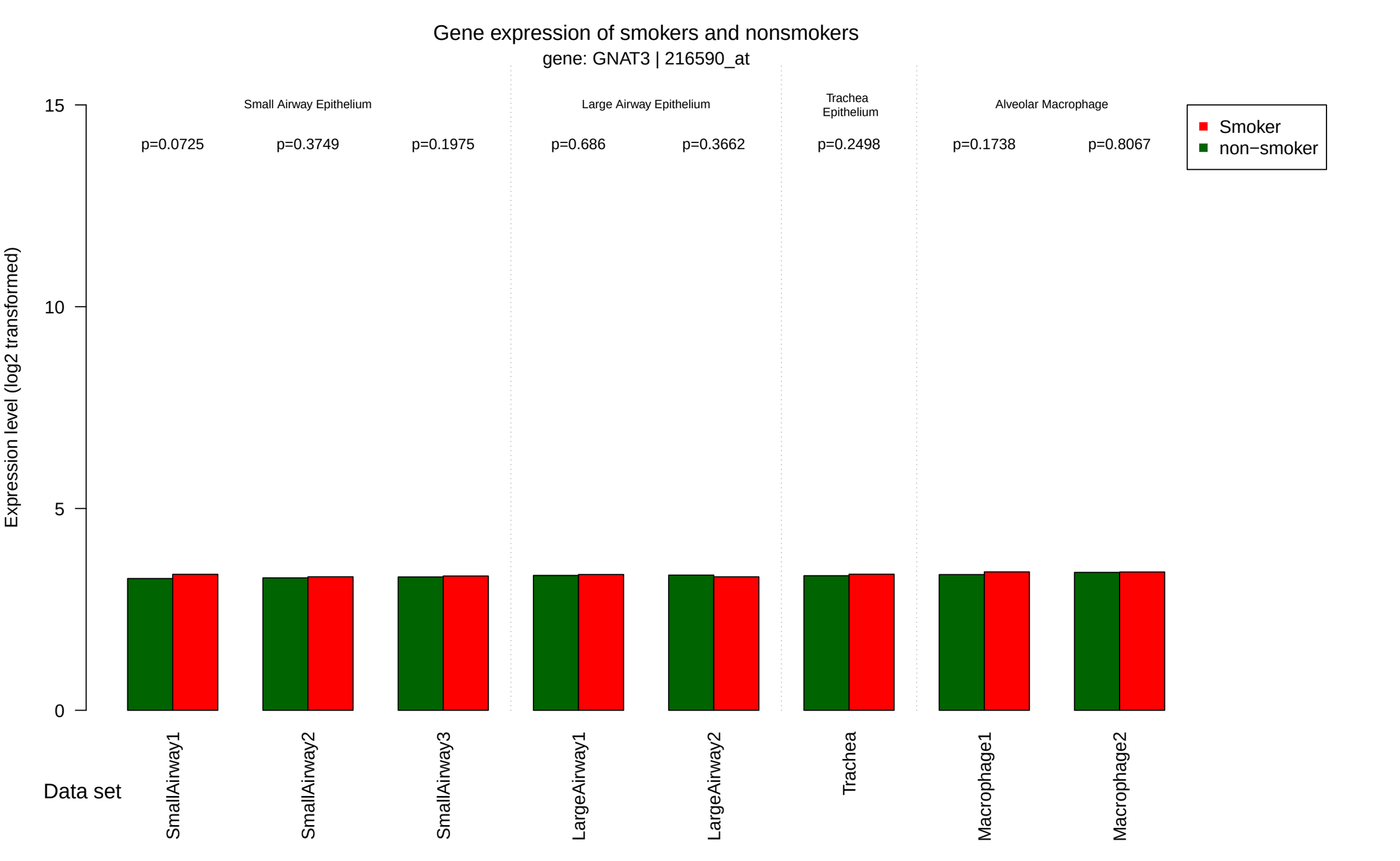
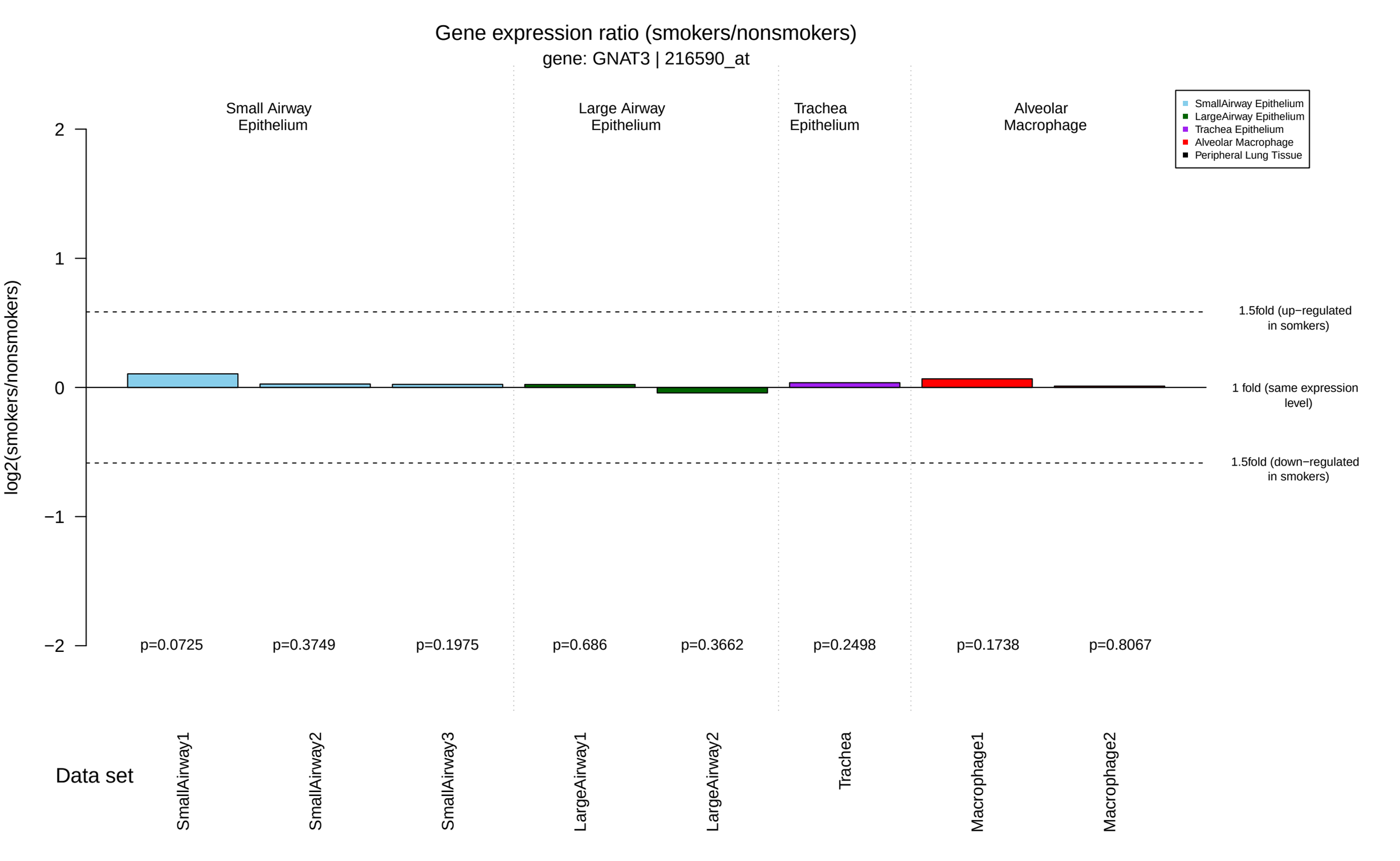
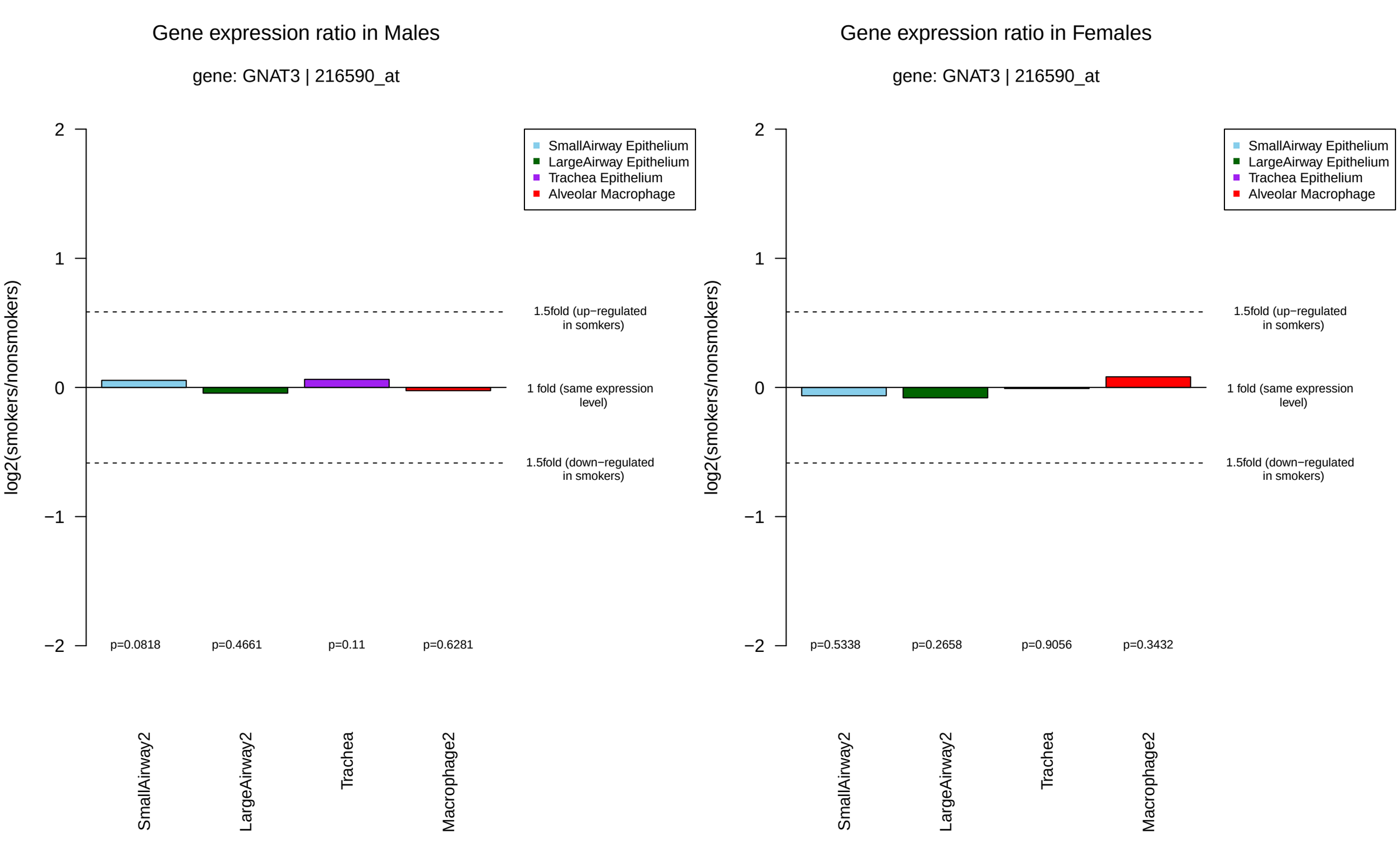
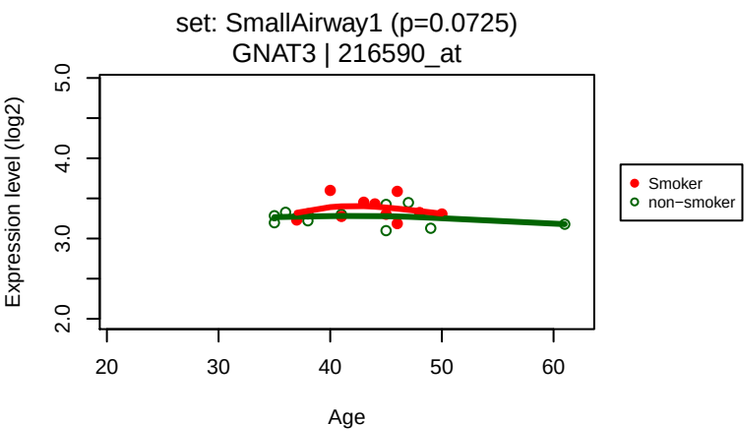
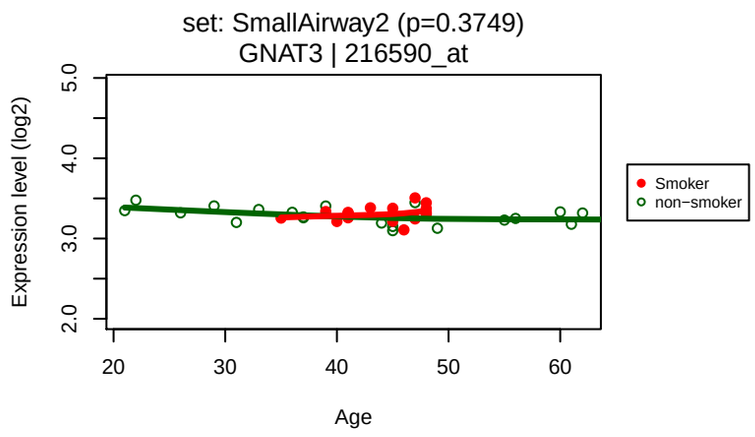
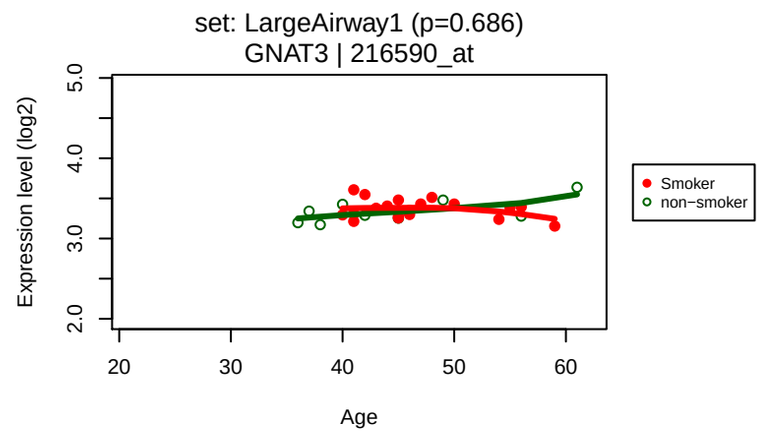

Gene expression of smokers and nonsmokers
gene: GNAT3 | 216590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0725
p=0.3749
p=0.1975
p=0.686
p=0.3662
p=0.2498
p=0.1738
p=0.8067
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

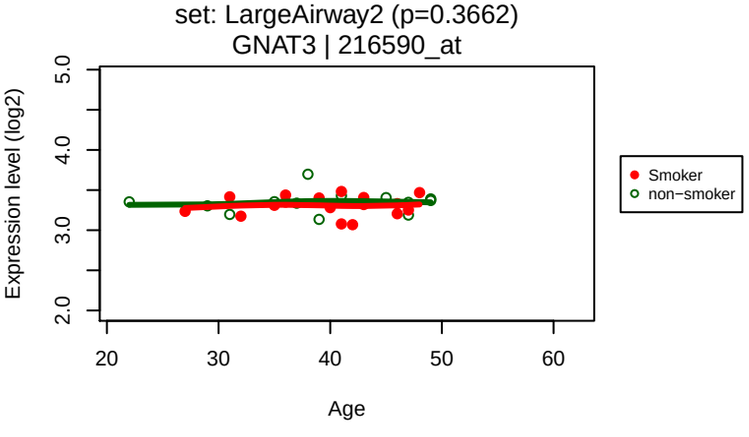

Gene expression ratio (smokers/nonsmokers)
gene: GNAT3 | 216590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0725
p=0.3749
p=0.1975
p=0.686
p=0.3662
p=0.2498
p=0.1738
p=0.8067
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

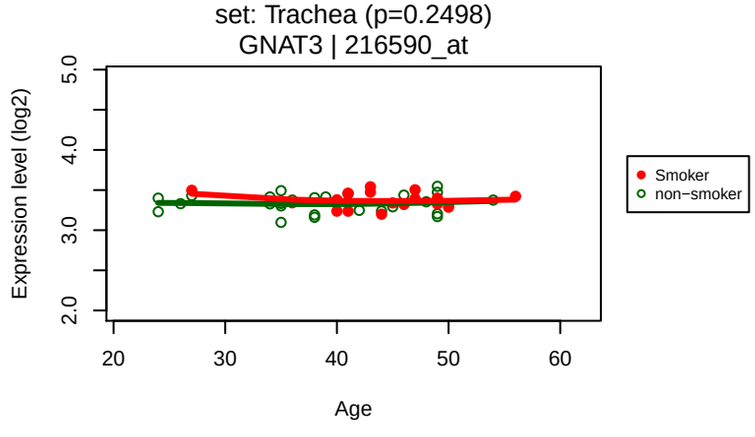

Gene expression ratio in Males
gene: GNAT3 | 216590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0818
p=0.4661
p=0.11
p=0.6281
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAT3 | 216590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5338
p=0.2658
p=0.9056
p=0.3432
SmallAirway2
LargeAirway2
Trachea
Macrophage2

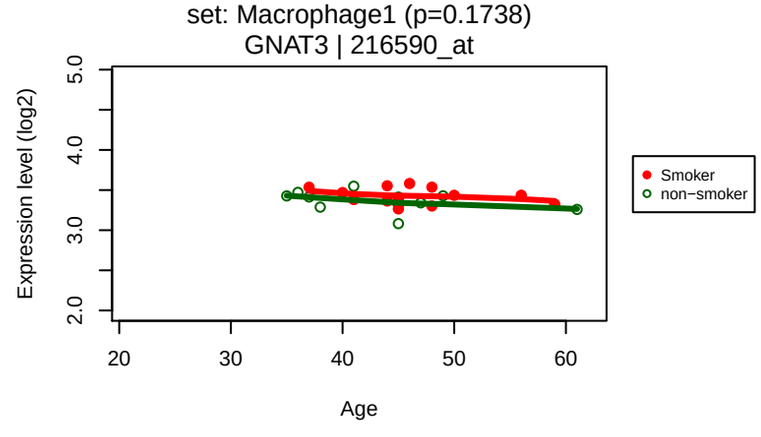

set: SmallAirway1 (p=0.0725)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker

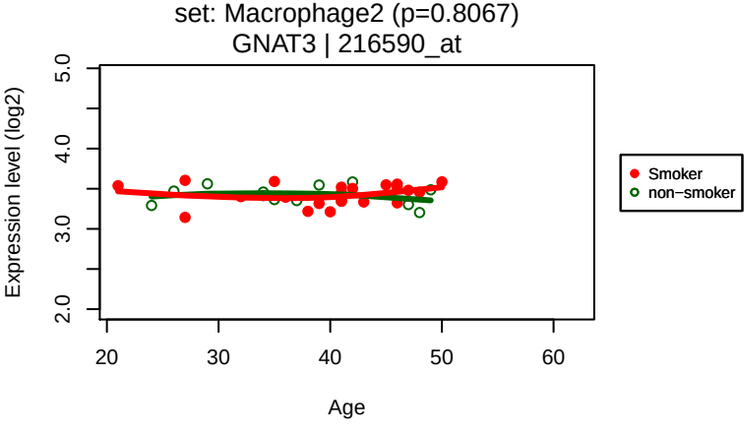

set: SmallAirway2 (p=0.3749)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker

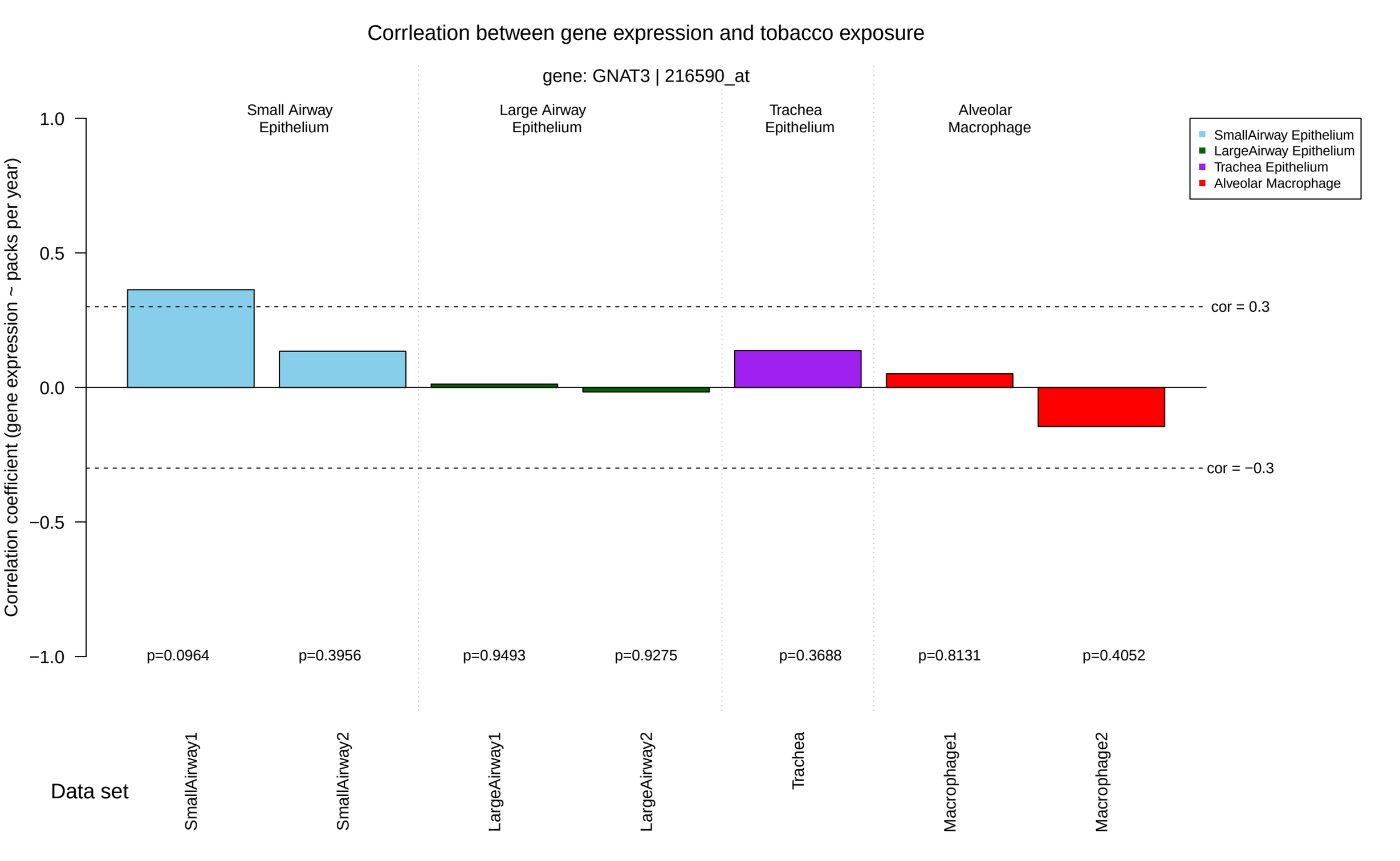

set: LargeAirway1 (p=0.686)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3662)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2498)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1738)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8067)
GNAT3 | 216590_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAT3 | 216590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0964
p=0.3956
p=0.9493
p=0.9275
p=0.3688
p=0.8131
p=0.4052
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2